Introduction to the Heisenberg Group

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§1 An Efficient Highway System

The matrix Heisenberg group $\mathbb{H}$ is the subgroup of $GL(3,\mathbb{R})$ consisting of matrices of the form

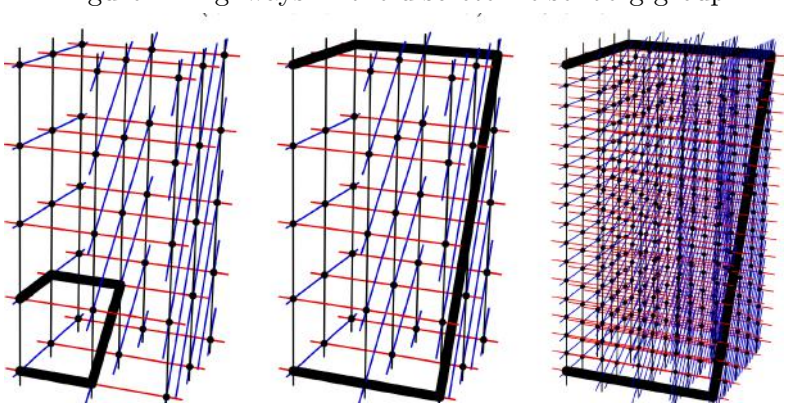
$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \qquad x, y, z \in \mathbb{R}.$$

This is a 3 dimensional Lie group that we can identify with $\mathbb{R}^3$. A simple calucation shows that the group law we get from matrix multiplication is given by

$$(x_1, y_1, z_1) \cdot (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2 + x_1 y_2) \tag{1}$$

In order to get a feel for what $\mathbb{H}$ really looks like, consider the group obtained by

Figure 1: Highways in the discrete Heisenberg group



restricting x, y, z to $\mathbb{Z}$. We can visualize[1] the group as a graph with vertex set $\mathbb{Z}^3$ where (x, y, z) is connected by an edge to the neighbors $(x \pm 1, y, z)$ and $(x, y \pm 1, z \pm x)$. We can always take the scaling limit of this graph to obtain the continuous group.

Figures 1 and 3 are courtesy of Robert Young.

[1] More precisely, we consider the Cayley graph of the generating set $S = \{(0, 0, \pm 1), (0, \pm 1, 0), (\pm 1, 0, 0)\}$

If we take the graph metric d_{graph} then interestingly we see

$$d_{graph}((x,y,z),(x,y,z+k)) \sim \sqrt{|k|}$$

For this reason, the late W. Thurston remarked that the discrete Heisenberg group was a design for an efficient highway system. We will see the same behaviour in the vertical direction with the continuous group.

§2 H & its Lie Algebra

More abstractly, we define the nth Heisenberg group $\mathbb{H}^n$ as the set $\mathbb{C}^n \times \mathbb{R}$ with the group operation given by

$$(\zeta, t) \cdot (\eta, s) = (\zeta + \eta, t + s + 2\,\mathrm{Im}(\zeta \cdot \bar{\eta})) \tag{2}$$

where $\zeta, \eta \in \mathbb{C}^n$ and $t, s \in \mathbb{R}$. This gives $\mathbb{C}^n \times \mathbb{R}$ a simple but interesting non-abelian group structure. For $n = 1$ this group is isomorphic to the matrix group by the assignment $(x, y, t) \to (x, y, t + 2xy)$, but the former presentation is generally preferred.

Exercise 2.1. Verify that the map φ into the matrix group given in (1) defined by

$$\varphi((x,y,t)) = \begin{pmatrix} 1 & x & t+2xy \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}$$

is a group isomorphism.

For the moment let us restrict our attention to $\mathbb{H}^1$. To compute its Lie Algebra $\mathfrak{h}$, we determine the left-invariant vector fields with respect the coordinate basis vectors. To that end, let $\gamma(s) = \tau s$ be the path whose tangent vector at the origin is $\tau \in \mathfrak{h}$.[2] We first compute the vector field X whose value at the origin is $\partial/\partial x$. Let $h = (\zeta, t) = (x + iy, t) \in \mathbb{H}^1$ and $\tau = (1, 0, 0)$. We test X on a smooth function $f \in C^\infty(\mathbb{H})$

$$\begin{aligned} X_h f = \mathrm{d}(L_h)_e(\gamma'(0))f &= \frac{\mathrm{d}}{\mathrm{d}s}\Big|_{s=0} f(h \cdot \tau s) \\ &= \frac{\mathrm{d}}{\mathrm{d}s}\Big|_{s=0} f(\zeta + (s,0,0), t + 2ys) = \frac{\partial f}{\partial x} + 2y\frac{\partial f}{\partial t}. \end{aligned}$$

We have shown

$$X = \frac{\partial}{\partial x} + 2y\frac{\partial}{\partial t}$$

Similarly we can compute the left invariant vector fields Y and T corresponding to $\tau = (0, 1, 0)$ amd $\tau = (0, 0, 1)$:

$$Y = \frac{\partial}{\partial y} - 2x\frac{\partial}{\partial t}, \qquad T = \frac{\partial}{\partial t}$$

The Lie brackets of these vector fields are $[Y, X] = 4T$, and all other brackets 0.

Exercise 2.2. Do the same computation to determine $\mathfrak{h}^n$, the Lie algebra of $\mathbb{H}^n$, and its Lie bracket relations.

[2] Here we are using the cannonical identification $\mathbb{R}^3 \cong T_0\mathbb{R}^3$

§2.1 Carnot Groups

The Lie algebra $\mathfrak{h}^n$ is in some sense as nice as possible. The vanishing of the brackets means the Lie algebra is (step-two) nilpotent. $\mathbb{H}^n$ is an example of a Carnot group, a simply connected nilpotent group whose Lie algebra $\mathfrak{g}$ admits a stratification, that is, there are linear subspaces $V_1, \dots V_r$ such that

$$\mathfrak{g} = V_1 \oplus V_2 \oplus \cdots \oplus V_r. \tag{3}$$

and $[V_1, V_{k-1}] = V_{k+1}$ for $k = 2, 3, \dots, r$ where we set $V_{r+1} = \{0\}$. The whole Lie algebra can then be generated by V_1, which is sometimes called the horizontal subspace.

$\mathfrak{h}$ has such a structure, for example. In fact, if we begin with a Lie algebra with a stratification $\mathfrak{h} = \mathfrak{h}_1 \oplus \mathfrak{h}_2$ with $\mathfrak{h}_1$ spanned by X and Y and $\mathfrak{h}_2$ spanned by T such that the only non-zero bracket is $[Y, X] = 4T$, then we can recover $\mathbb{H}$ via the exponential map, $\exp : \mathfrak{h} \to \mathbb{H}^1$. By the Baker-Campbell-Hausdorff formula we have

$$\exp(X_1)\exp(X_2) = \exp\left(X_1 + X_2 + \frac{1}{2}[X_1, X_2]\right).$$

We can use exponential coordinates $(x, y, t) \leftrightarrow \exp(x \cdot X + y \cdot Y + t \cdot T)$ to deduce the original group law.

§2.2 Contact structure

The subspace spanned by X and Y determines a horizontal distribution, that is, a rank 2 subbundle of TM. We denote the horizontal distribution as $T\mathbb{H}$. In other words, at each tangent space $T_p\mathbb{H}$ there is a 2 dimensional subspace $H_p\mathbb{H}$. This plane is called the point of contact. The contact structure is best described in the language of differential forms.

Definition 2.3. A contact form on a $2n+1$ dimensional manifold M is a non-vanishing differential 1-form ω with the property that at each point $p \in M$ the restriction of $\mathrm{d}\omega_p$ to $\ker \omega_p$ is nondegenerate.

In the case of $\mathbb{H}$ we take

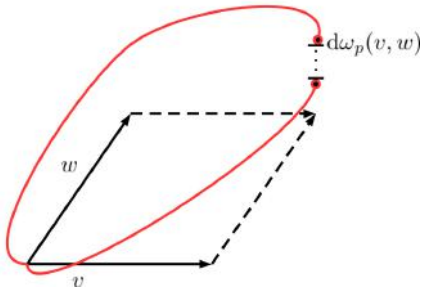
$$\omega = -2y\mathrm{d}x + 2x\mathrm{d}y + \mathrm{d}t.$$

Exercise 2.4. Show that $\ker \omega_p = \mathrm{span}\{X_p, Y_p\}$ for all $p \in \mathbb{H}$.

The reader may notice that $f\omega$ determines the same contact structure for any $f \in C^\infty(\mathbb{H})$. However, the contact form is actually unique up to a change of coordinates by (the contact version of) Darboux's theorem.

The non-integrability condition ensures that there is no integral manifold of the horizontal distribution, or in the case of $\mathbb{H}$, a surface whose tangent plane at every point is given by $\ker \omega_p$. The non-existence of an integral manifold is where the interesting geometry of $\mathbb{H}$ arises from.

Figure 2: Integral curves of $H\mathbb{H}$ are shown in red



Heuristically, we can consider two tangent vectors $v, w \in H_p\mathbb{H}$ and look at the two integral curves starting the origin and following the parallelogram spanned by v and w. By Stoke's theorem, the 2 form defined by the signed vertical distance between the endpoints of the integral curves agrees with $\mathrm{d}\omega_p(v,w)$. In other words, nonintegrability requires that these integral curves do not close up an form a surface.

Contact geometry is closely related to symplectic geometry; indeed the condition on $\mathrm{d}\omega$ is equivalent to saying that it is a symplectic form. Furthermore, we see the group law (2) can be written in terms of the cannonical symplectic form on $\mathbb{R}^{2n}$, i.e.

$$\operatorname{Im} \zeta \cdot \bar{\eta} = S(\zeta, \eta) = \zeta_1 \cdot \eta_2 - \zeta_2 \cdot \eta_1$$

where $\zeta = \zeta_1 + i\zeta_2$, $\eta = \eta_1 + i\eta_2$. Thus any symplectomorphism σ yields an automorphism of $\mathbb{H}^n$, $(\zeta, t) \mapsto (\sigma(\zeta), t)$. In fact, we can consider $\mathfrak{h}^n$ as a central extension of the (commutative) Lie algebra $(\mathbb{R}^{2n}, S)$ by $\mathbb{R}$, which is expressed by the following exact sequence:

$$0 \to \mathbb{R} \to \mathfrak{h}^n \to \mathbb{R}^{2n} \to 0.$$

§3 Sub-Riemannian Structure

We can put a metric on $\mathbb{H}$ in a similar fashion to a Riemannian metric: we infinimize length of curves, but by considering only paths whose tangent vectors are in the horizontal subspace.

Definition 3.1. A horizontal curve or *Legendrian curve* is an absolutely continuous curve $\gamma : [0,1] \to \mathbb{H}$ such that $\omega(\gamma'(t)) = 0$ for all $t \in [0,1]$.

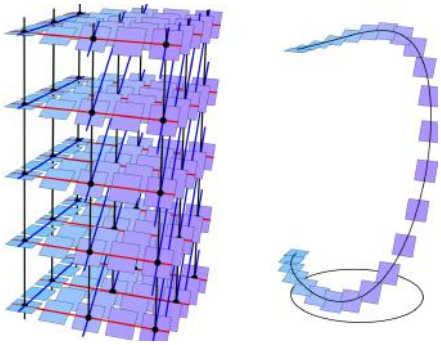


Figure 3: (left) The contact distribution at several points. (right) A Legendrian curve

§3.1 Metric

We first put an inner product $\langle \cdot, \cdot \rangle$ on $H\mathbb{H}$ by declaring X and Y to be orthonormal. Let $|\cdot|$ denote the induced norm. Now we can define the *Carnot Carathéodory distance*:

$$d_{CC}(p,q) = \inf \int_0^1 |\gamma'(t)| \mathrm{d}t, \qquad p, q \in \mathbb{H}$$

where the infimum is taken over all Legendrian curves with $\gamma(0) = p$ and $\gamma(1) = q$. Note that this length is the same as the length of $\pi(\gamma)$, the projection of γ onto $\mathbb{C}$. This distance is defined for all points in $\mathbb{H}$ thanks to the following theorem:

Theorem 3.2 (Chow's Theorem)

Let G be a Carnot group. Then any two points $p, q \in G$ can be joined by a minimal Legendrian curve. Thus d_{CC} determines a left-invariant metric.

Proof. (for $\mathbb{H}$) Consider the action by right multiplication of the one-parameter subgroups generated by X and Y. The orbits of these subgroups are clearly tangent to $H\mathbb{H}$. On the other hand, X and Y generate $\mathfrak{h}$ so the corresponding subgroups generate $\mathbb{H}$. Thus every pair of points in $\mathbb{H}$ can be joined by a piecewise smooth curve whose segments lie in the orbits of $R_{\exp(tX)}$ or $R_{\exp(tY)}$. □

Geodesics of this metric travelling vertically are arcs of helices (see Figure 2). In fact, the vertical rise of a Legendrian curve γ is proportional to the signed area of the convex hull of the projection $\pi(\gamma)$, where π is the projection onto $\mathbb{C}$. To see this, suppose $\gamma = (\gamma_1, \gamma_2, \gamma_3)$ and $\gamma(0) = 0$, $\gamma(1) = p = (\zeta, t)$. Then we have

$$0 = \omega(\gamma') = -2\gamma_2\gamma_1' + 2\gamma_1\gamma_2' + \gamma_3'$$
$$\implies \gamma_3 = \int_0^1 2(\gamma_2\gamma_1' - \gamma_1\gamma_2') = -4\int_\Omega \mathrm{d}x \wedge \mathrm{d}y$$

where $\Omega \subset \mathbb{C}$ is the region bound by $\pi(\gamma)$. Minimizing the length of a curve whose convex hull has a fixed area is closely related to the classical isoperimetric problem. Indeed, the

minimum length is obtained when $\pi(\gamma)$ is the arc of a circle. When $p = (0, t)$, $\pi(\gamma)$ is a circle. Note that

$$d((x, y, t), (x, y, t + k)) \sim \sqrt{|k|}$$

just as in the discrete group.
There is also the *Koranyí guage* defined by

$$||p|| = (|z|^4 + t^2)^{1/4}$$

where $p = (z, t) \in \mathbb{H}$. From this we can construct the *Koranyí metric*

$$d(p, q) = ||q^{-1} \cdot p||.$$

which is clearly left-invariant. Similar constructions in Carnot groups are typically only psuedometrics, but in this case one can check that the triangle inequality holds. It turns out this metric is equivalent to the Carnot-Carathéodory metric.

Exercise 3.3. Show that $d(p, q) = d_{CC}(p, q)$.

§3.2 Homogeneous and Hausdorff Dimension

A nice property of the Koranyí metric is that it comes with a family of (non-isotropic) dilations $\delta_s(z, t) = (sz, s^2 t)$, where $s \in \mathbb{R}$. This makes $\mathbb{H}$ into a homogeneous space.

Definition 3.4. Let G be a Carnot group with a Lie algebra as in (2) and let $d_i = \dim \mathfrak{g}_i$. The *homogeneous dimension* of G is the number

$$Q = d_1 + 2d_2 + \cdots + rd_r$$

Thus the homogeneous dimension of $\mathbb{H}^n$ is $2n + 2$. The name comes from the fact that when a ball is dilated by r, the volume dilates by a factor of r^{2n+2}.

A more common notion of dimension we can also compute is Hausdorff dimension. Recall the α-Hausdorff dimension is defined as

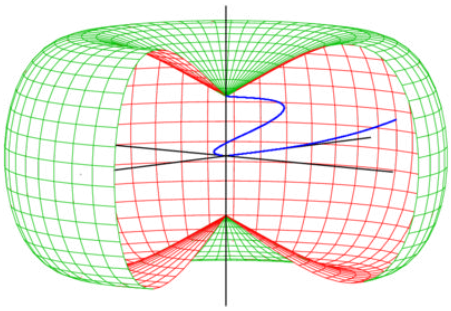
$$\mathcal{H}^\alpha(A) = \lim_{\delta \to 0} \inf_{\mathcal{B}} \sum_{B \in \mathcal{B}} (\operatorname{diam} B)^\alpha$$

where the infinimum is over covers of A by (closed) balls of diameter at most δ. Note that $\mathcal{H}^\alpha(L_h E) = \mathcal{H}^\alpha(E)$ and $\mathcal{H}^\alpha(\delta_s E) = s^\alpha \mathcal{H}^\alpha(E)$. Hence there exists a consant $C_\alpha \in [0, \infty]$ such that

$$\mathcal{H}^\alpha(B(p, r)) = C_\alpha r^\alpha$$

where $B(p, r) = \{q \in \mathbb{H} : d(p, q) < r\}$. One can check using the Koranyí norm that $\alpha = Q = 2n + 2$ is the Hausdorff dimension of $(\mathbb{H}^n, d_{CC})$, i.e. $C_Q < \infty$ and $C_\alpha = 0$ for all $\alpha < Q$. Thus $\mathbb{H}^n$ qualifies as a fractal in the sense that its Hausdorff dimension exceeds its topological dimension.
The remainder of this paper is dedicated to surveying some areas of mathematics and

Figure 4: Cross section of a Heisenberg ball centered at the origin.



physics where $\mathbb{H}^n$ shows up. We do not make any claims of being comprehensive, but will merely highlight a few relevant topics.

§4 Geometry of Several Complex Variables

Consider the upper half space of $\mathbb{C}^{n+1}$

$$\mathcal{U}^n = \left\{ z = (z_1, \ldots, z_{n+1}) \in \mathbb{C}^{n+1} : \operatorname{Im}(z_{n+1}) > \sum_{j=1}^{n} |z_j|^2 \right\}$$

In the case of $\mathbb{C}$, $\mathcal{U}$ has a rich group of symmetries given by the special linear group $SL_2(\mathbb{R})$. In higher dimensions, $\mathbb{H}^n$ describes the translational symmetry of $\mathcal{U}^n$. More precisely, $(\zeta, t) \in \mathbb{H}^n$ acts on $\mathcal{U}^n$ by the action

$$(\zeta, t) \cdot (z', z_{n+1}) = (z' + \zeta, t + z_{n+1} + 2iz' \cdot \bar{\zeta} + i|\zeta|^2).$$

The transformation preserves the "radius" $r(z) = \operatorname{Im} z_{n+1} - |z'|^2$ since

$$|z' + \zeta|^2 - |z'|^2 = \operatorname{Im} i(z'\zeta + |\zeta|^2)$$

so that $\mathcal{U}^n = \{z : r(z) > 0\}$ and $\partial\mathcal{U}^n = \{z : r(z) = 0\}$ are preserved under the action.

Exercise 4.1. Verify that the above transformation determines a group action.

The action is faithful as well as simply transitive on $\partial\mathcal{U}$, that is, for any $w, z \in \partial\mathcal{U}$ there is a unique group element that sends w to z. We may therefore identify $\mathbb{H}^n$ and $\partial\mathcal{U}$ by its action on the origin

$$(\zeta, t) \longleftrightarrow (\zeta, t) \cdot (0, 0) = (\zeta, t + i|\zeta|^2)$$

§4.1 CR Structures

A Cauchy-Riemann (CR) structure provides a way to describe hypersurfaces in $\mathbb{C}^n$ by studying its tangent space. Again we will consider the simplest case, but the results are easily generalizable.

Definition 4.2. A 3 dimensional CR manifold is a real, oriented 3-manifold M endowed with a subbundle $T^{1,0}$ of the complexified tangent bundle $TM \otimes \mathbb{C}$ such that $\dim_{\mathbb{C}} T^{1,0} = 1$, $T^{1,0} \cap \bar{T}^{1,0} = \{0\}$ and $T^{1,0}$ is integrable.

We will write $\bar{T}^{1,0} = T^{0,1}$. To furnish $\mathbb{H}$ with a CR structure, we complexify our basis vector fields by introducing

$$\bar{Z} = \frac{1}{2}(X - iY) = \frac{\partial}{\partial \bar{z}} - iz\frac{\partial}{\partial t}$$
$$Z = \frac{1}{2}(X + iY) = \frac{\partial}{\partial z} + i\bar{z}\frac{\partial}{\partial t}$$

where

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right), \quad \frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right).$$

Exercise 4.3. Show that $[\bar{Z}, Z] = 2iT$ and $Z \cap \bar{Z} = \{0\}$. In particular, $\mathbb{H}$ is a CR-manifold with $T^{1,0} = \operatorname{span} Z$.

We will see that the CR structure on $\mathbb{H}$ is the same as the restriction of the complex structure on $\mathbb{C}^2$. To be specific, for coordinates (z_1, z_2) we have

$$T^{1,0}\mathbb{C}^2 = \operatorname{span}\left\{\frac{\partial}{\partial z_1}, \frac{\partial}{\partial z_2}\right\}, \quad T^{0,1}\mathbb{C}^2 = \operatorname{span}\left\{\frac{\partial}{\partial \bar{z}_1}, \frac{\partial}{\partial \bar{z}_2}\right\}.$$

Vector fields in $T^{0,1}\mathbb{C}^2$ are called Cauchy Riemmann operators because they are precisely the operators that annihilate holomorphic functions.

The (holomorphic) tangent space at a point of ∂U consists of vectors in $T^{0,1}$ that annihilate $r(z)$ whenever $r(z) = 0$. It is obvious from the definition that $\bar{Z}$ and Z are tangential. $\bar{Z}$ is clearly in $T^{1,0}\mathbb{C}^2$ at the origin of $\mathbb{H}$. $\bar{Z}$ is left invariant and acts holomorphically on $\partial\mathcal{U}^2$, and so preserves $T^{1,0}\mathbb{C}^2$. Using a similar argument for Z, the CR structures of $\partial\mathcal{U}^2$ and $\mathbb{H}$ coincide everywhere.

Exercise 4.4. Generalize the above to $\partial\mathcal{U}^n$ and $\mathbb{H}^n$.

§5 Quantum Mechanics and Unitary Representation

We delineate the relationship of $\mathbb{H}^n$ to quantum mechanics. Indeed, the reader may have noticed that "Heisenberg group" is an eponym of the famous physicist[3].

Classically, the position and momentum of a particle

$$q = (q_1, \ldots, q_n), \quad p = (p_1, \ldots, p_n) \in \mathbb{R}^n$$

[3]Physicists call it the Weyl group

are points of the phase space. In quantum mechanics, we replace these with the position and momentum as operators $Q_j = x_j$ and $P_j = -i\hbar\partial/\partial x_j$ on the complex Hilbert space $\mathcal{H} = L^2(\mathbb{R}^n)$ that satisfy the so-called canonical commutator relations

$$[P_j, Q_k] = -i\hbar\delta_{jk}I \tag{4}$$

where $\hbar$ is Planck's constant and I is the identity. For conenienve we choose units so that $\hbar = 1$. We wish to associate to each classical observable (continuous function of p and q) a corresponding operator on $\mathcal{H}$. This process is known as *quantization*. There are many different quantization schemata but the one we will focus on is *Weyl quantization*. First suppose

$$a(p, q) = e^{i(u\cdot q + v\cdot p)} \qquad u, v \in \mathbb{R}^n.$$

The idea is to quantize operators of this form, then quantize a general observable by Fourier transformation. We associate to a the operator

$$W(u, v) = e^{i(u\cdot Q + v\cdot P)}$$

called the *Weyl correspondence*, which is uniquely characterized by

$$(W(u, v)f)(x) = e^{ix\cdot u}e^{iu\cdot v/2}f(x + v), \quad f \in \mathcal{H}.$$

This is a restriction of a unitary representation (see 6.2): for each $(\zeta, t) = (u + iv, t) \in \mathbb{H}^n$ define

$$(R(\zeta, t)f)(x) = e^{i(u\cdot x + u\cdot v/2 + t/4)}f(x + v).$$

Exercise 5.1. Show that $R : \mathbb{H}^n \to U(\mathcal{H})$ is a homomorphism, where $U(\mathcal{H})$ is the group of unitary operators.

More generally, for $\lambda \in \mathbb{R}$ define

$$(R^\lambda(\zeta, t)f)(x) = e^{i\lambda(u\cdot x + u\cdot v/2 + t/4)}f(x + v). \tag{5}$$

This corresponds to scaling Planck's constant by $1/\lambda$. We can obtain the corresponding representation of $\mathfrak{h}^n$ by differentiation. We obtain

$$\frac{\partial R^\lambda}{\partial u^i}(0) = i\lambda Q_i, \qquad \frac{\partial R^\lambda}{\partial t}(0) = \frac{i}{4}\lambda I$$
$$\frac{\partial R^\lambda}{\partial v^j}(0) = i\lambda P_j.$$

which are the representations of X_i, T, and Y_j respectively.

Lemma 5.2

The representations R^λ are irreducible and faithful.

Proof. Since R^λ is unitary, it suffices to show that if A is a bounded linear operator that commutes with $R^\lambda(\zeta, t)$ for all $(\zeta, t) \in \mathbb{H}^n$, then A is a mutiple of the identity. In particular, such an operator commutes with translations i.e. (5) with $u = t = 0$. By the theory of Fourier analysis we see A is a Fourier multiplier: there exists a bounded function $a : \mathbb{R}^n \to \mathbb{R}$ such that $\widehat{A\psi(\xi)} = a(\xi)\hat{\psi}(\xi)$. Taking (5) with $v = t = 0$ we see $a(\xi + u) = a(\xi)$, so a is constant. It's clear that R^λ and R^η are distinct when $\eta \neq \lambda$ by setting $\zeta = 0$ $\square$

It turns out that these are the *only* irreducible representations of the Heisenberg algebra.

Theorem 5.3 (Stone-von Nuemann Theorem)

The only non-trivial irreducible representations of $\mathbb{H}^n$ are $\{R^\lambda,\ \lambda \in \mathbb{R}\}$, up to unitary equivalence.

As infinite-dimensional representations get thorny, we will not go into the proof here. The upshot of the theorem is that once Planck's constant is fixed, there is a unique irreducible representation of $\mathfrak{h}^n$, which is why physicists can calculate using the choice of Q_i and P_j together with the canonical commutation relations (4) without worry.

§6 Appendix

We state standard material, and refer the reader to any textbook on manifolds and Lie groups.

§6.1 Lie Group Theory

Definition 6.1. A Lie Group is a smooth manifold G that is equipped with a group structure. We require multiplication $m : G \times G \to G$ and inversion $i : G \to G$ given by

$$m(g, h) = g \cdot h, \quad i(g) = g^{-1}$$

to be smooth maps.

Definition 6.2. A Lie Algebra is a vector space $\mathfrak{g}$ equipped with an alternating, anti-symmetric bilinear form $[\cdot, \cdot]$ that satisfies the Jacobi identity:

$$[x, [y, z]] + [z, [y, x]] + [y, [x, z]] = 0$$

for all $x, y, z \in \mathfrak{g}$.

A basic example is the commutator bracket on $\mathfrak{X}(M)$, the smooth vector fields on M. It is defined by $[X, Y] = XY - YX$.

Definition 6.3. The Lie algebra $\mathfrak{g}$ of a Lie group G the set of all *left-invariant* vector fields $X \in \mathfrak{X}(M)$, i.e. $(L_g)_*(X) = X$ for all $g \in G$, where L_g denotes the left multiplication map. This amounts to

$$\mathrm{d}(L_g)_h(X_h) = X_{gh}, \quad g, h \in G$$

where d denotes the tangent map. $\mathfrak{g}$ is bijective with the tangent space at the identity, for any $\xi \in T_eG$ corresponds to the left invariant vector field ξ^L defined by

$$\xi^L|_g = d(L_g)_e(\xi)$$

The *raison d'etre* of the Lie algebra is to study the "infinitesimal action" of the Lie group. We can go from the Lie algebra via the exponential map.

Definition 6.4. A Lie aglebra homomorphism is

Definition 6.5. Let $\xi \in \mathfrak{g}$. The one-parameter subgroup corresponding to ξ is the unique solution to the inital value problem

$$\gamma'(t) = \xi^L, \quad \gamma(0) = e$$

The exponential map of ξ is defined as

$$\exp(\xi) = \gamma(1)$$

The exponential map is named as such because in the case of a matrix Lie group, it coincides with the matrix exponential.

Example 6.6

The group of rotation matrices $SO(3) = SL(3, \mathbb{R}) \cap O(3)$ consists of orthogonal matrices with determinant 1. The Lie algebra $\mathfrak{so}(3)$ consists of the 3×3 skew symmetric matrices.

§6.2 Representations

Definition 6.7. A representation of a group G on a vector space V over the field F is a group homomorphism $R : G \to GL(V, F)$. Similarly, a representation of a Lie algebra $\mathfrak{g}$ is a Lie algebra homomorphism $R : \mathfrak{g} \to \mathfrak{gl}(V, F)$, that is, a linear map that preserves the Lie bracket.

Definition 6.8. A *unitary representation* of a group G on a Hilbert space $\mathcal{H}$ is a representation on the group of unitary operators $R : G \to U(\mathcal{H})$ such that $g \mapsto \varphi(g)\psi$ is smooth in the norm topology for all $\psi \in \mathcal{H}$.

Definition 6.9. Two representations R_1 and R_2 of G are said to be *unitarily equivalent* if there is a unitary operator U such that

$$U \circ R_1(g) = R_2(g) \circ U, \quad \forall\, g \in G$$

Definition 6.10. A representation R of G on V is said to be *irreducible* if there are no non-trivial subspaces of V that are invariant under the induced action of G on V. It is said to be *faithful* if R is injective.